

## WITHDRAWN (2026-05-17)

Observation 1 of this combined note is **false**. At  $t = 5$ , a planar graph of Voigt (1993) has  $\chi_\ell(G) = 5$  (every planar graph is 5-choosable by Thomassen 1994; Voigt's example is not 4-choosable) and  $\text{cr}(G) = 0 < 1 = \text{cr}(K_5)$ , which directly falsifies the claim " $\chi_\ell(G) \geq t$  and  $t \leq 18 \Rightarrow \text{cr}(G) \geq \text{cr}(K_t)$ ". Independently, the proof in this note appeals to the Albertson–Cranston–Fox / Barát–Tóth / Ackerman chain under the (incorrect) belief that the chain uses  $\chi$  only through Dirac's  $\delta \geq t - 1$  bound; Ackerman ([arXiv:1509.01932](#), §3.1) in fact uses the minimum edge-count function  $f_r(n)$  for  $r$ -critical graphs, and the list-critical edge floor (Krivelevich 1997) is strictly weaker, so the lift is unjustified even apart from the  $t = 5$  counterexample.

Observation 2 (the bisection-width Crossing Lemma for regular spectral expanders) is salvageable, but the version here contains several fixable but real errors: an odd- $n$  floor in Alon's spectral bisection bound, a wrong rounding bookkeeping step in Lemma 21, a numerical illustration below the Bungener–Kaufmann density threshold, and an overstated Ore-scope claim in the limitations section. The corrected stand-alone spectral note is at `D16_expander_crossing_paper/expander_crossing_v2.tex` (see `D17_submission_packets/paper_D16.md` for the current submission packet).

This combined-paper file is preserved for historical record only. The companion list-coloring source `D15_list_albertson_paper/list_albertson_le_18.tex` is similarly withdrawn.

# Two structural observations in the neighbourhood of Albertson’s conjecture

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## Abstract

We record two observations arising from an analysis of the structural slacks in the chain of partial results towards Albertson’s conjecture. **First**, the Albertson–Cranston–Fox / Barát–Tóth / Ackerman chain establishing the conjecture unconditionally at chromatic number up to  $t = 18$  depends on its  $\chi$ -hypothesis only through the Dirac minimum-degree bound, which lifts verbatim to list-coloring via the list versions of Dirac’s and Brooks’ theorems (Borodin 1977; Erdős–Rubin–Taylor 1979). Assembling these observations gives list-Albertson at  $t \leq 18$ , strictly stronger than ordinary Albertson at the same range. **Second**, the Pach–Spencer–Tóth bisection-width Crossing Lemma, combined with Alon’s spectral bisection lower bound, yields an explicit- $\theta$  Crossing Lemma for  $d_0$ -regular spectral  $\theta$ -expanders,

$$\text{cr}(G) \geq \frac{(1 - \theta)^2 d_0^2 n^2}{1280} - \frac{d_0^2 n}{16},$$

with an Albertson-type corollary on regular spectral-expander critical graphs. We are not aware of an earlier Crossing-Lemma inequality with explicit spectral dependence. Neither observation closes the Cranston residual at  $t \in \{25, 26\}$ ; both are recorded as side-observations in the structural neighbourhood of the conjecture.

## 1 Introduction

Albertson’s conjecture [3] is the natural common strengthening of the Four Colour Theorem and Guy’s formula for  $\text{cr}(K_n)$ :

**Conjecture 1** (Albertson). Every graph  $G$  with  $\chi(G) \geq t$  satisfies  $\text{cr}(G) \geq \text{cr}(K_t)$ .

The conjecture is currently known unconditionally for  $t \leq 24$  through a chain of partial results that all proceed by analysing a minimum counterexample (MCE). Each step combines two ingredients:

1. an *edge floor* on critical graphs –  $|E(G)| \geq \frac{k-1}{2}|V(G)|$  via Dirac’s theorem on critical graphs [14];
2. a *Crossing Lemma* input bounding  $\text{cr}(G)$  from below in terms of  $|E(G)|^3/|V(G)|^2$  once  $|E(G)|/|V(G)|$  exceeds an explicit threshold.

Together these inputs force a vertex bound  $|V(G)| \leq ct$  on any hypothetical MCE; the small remaining range is then closed either hand-by-hand or by a topological argument on graphs with few crossings per edge. The chain proceeds in four steps: Albertson–Cranston–Fox [4] (2010,  $t \leq 12$  at  $|V| \leq 4t$ ), Barát–Tóth [7] (2010,  $t \leq 16$  at  $|V| \leq 3.57t$ ), Ackerman [1] (2019,  $t \leq 18$  at  $|V| \leq 3.03t$  via  $|E| \leq 6n - 12$  for graphs with at most four crossings per edge), and Cranston [12] (2025,  $t \leq 24$  via a new structural lemma due to Fox–Pach–Suk [17]).

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**Two structural observations.** This note records two observations on the structural neighbourhood of this chain. Each is a "we observe that an existing chain of theorems lifts or packages to one degree of generality higher" statement; neither closes any open case of Conjecture 1.

**The list-coloring variant.** A *list assignment* on  $G$  is a function  $L : V(G) \rightarrow 2^{\mathbb{N}}$ ; a proper  $L$ -colouring is a proper colouring  $c$  with  $c(v) \in L(v)$  for every  $v$ . The graph  $G$  is  $L$ -colourable if such a colouring exists, and  $k$ -*list-colourable* (or  $k$ -*choosable*) if it is  $L$ -colourable for every  $L$  with  $|L(v)| \geq k$  for all  $v$ . The *list chromatic number*  $\chi_\ell(G)$  is the smallest such  $k$ . The notion is due to Vizing [35] and to Erdős, Rubin and Taylor [16], and satisfies  $\chi_\ell(G) \geq \chi(G)$  for every graph  $G$ . The list version of Albertson's conjecture is correspondingly stronger:

**Conjecture 2** (List-Albertson). Every graph  $G$  with  $\chi_\ell(G) \geq t$  satisfies  $\text{cr}(G) \geq \text{cr}(K_t)$ .

Despite the conceptual closeness of the two conjectures, the list-coloring variant appears to have received little attention in the literature; for small  $t \leq 4$  it reduces to elementary observations (any  $G$  with  $\chi_\ell(G) \geq 3$  is non-planar by the list version of the Four Colour Theorem for planar graphs – in fact every planar graph is 5-choosable, by Thomassen [34]). We are not aware of a result placing list-Albertson at a threshold higher than  $t = 4$ .

**The spectral Crossing-Lemma variant.** On the topological side, the standard Crossing Lemma of Ajtai, Chvátal, Newborn, Szemerédi [2] and Leighton [22] bounds  $\text{cr}(G)$  from below by  $c|E(G)|^3/|V(G)|^2$  once  $|E(G)| \geq 4|V(G)|$ ; successive improvements due to Pach–Tóth [30], Pach–Radoičić–Tardos–Tóth [31], Ackerman [1], and most recently Bungener and Kaufmann [11] (at  $c = 1/27.48$ ) have driven the constant down. Parallel to this density-iteration line, Leighton [23] introduced a bisection-width variant; Pach–Spencer–Tóth [29] (via the dual estimate of Pach–Shahrokhi–Szegedy [28] and Sýkora–Vrt’o [33]) established

$$\text{cr}(G) \geq \frac{\text{bw}(G)^2}{40} - \frac{\sum_v \deg(v)^2}{16}, \quad (1)$$

which we work with throughout in its self-contained  $1/80$  form (see Section 3.2). Combining (1) with Alon's spectral bisection bound for regular spectral expanders gives an explicit dependence of the Crossing-Lemma constant on the spectral gap parameter  $\theta$ .

## 1.1 Statements

The two main theorems of this note are the following.

**Theorem 3** (Observation 1: list-Albertson at  $t \leq 18$ ). *Let  $G$  be a graph and let  $t \leq 18$  be an integer. If  $\chi_\ell(G) \geq t$ , then  $\text{cr}(G) \geq \text{cr}(K_t)$ .*

**Theorem 4** (Observation 2: bisection-width Crossing Lemma for regular spectral expanders). *Let  $G$  be a  $d_0$ -regular graph on  $n$  vertices with second adjacency eigenvalue  $|\lambda_2(G)| \leq \theta d_0$  for some  $\theta \in [0, 1)$ . Then*

$$\text{cr}(G) \geq \frac{(1 - \theta)^2 d_0^2 n^2}{1280} - \frac{d_0^2 n}{16}. \quad (2)$$

*The constant  $1/1280$  can be replaced by  $1/640$  if one invokes the sharper Pach–Spencer–Tóth form of (1) with constant  $1/40$  in place of  $1/80$  (Remark 22).*

Theorem 3 is strictly stronger than the corresponding unconditional case of Albertson's conjecture (namely Conjecture 1 at  $t \leq 18$ , established by [1]) because the hypothesis  $\chi_\ell(G) \geq t$  is weaker than  $\chi(G) \geq t$ . Theorem 4 is, to our knowledge, the first explicit packaging of the spectral parameter  $\theta$  inside a Crossing-Lemma constant; the underlying ingredients (the PST inequality and the expander mixing lemma) are classical.

**Honesty note.** Both observations recorded here are not so much new theorems as observations that existing chains of theorems lift or package to one degree of generality higher. We have not, however, found either assembly in the literature, and the quantitative thresholds ( $t \leq 18$  for Theorem 3, the explicit- $\theta$  form of (2) for Theorem 4) do not appear to have been previously recorded. We frame the contribution accordingly.

**Organisation.** Section 2 establishes Theorem 3: it fixes notation, states the two list-coloring inputs (list versions of Dirac and of Brooks), assembles the list-ACF / list-BT / list-Ackerman chain, explains why the same lift currently *stops* at  $t = 18$ , and records a DP-coloring corollary. Section 3 establishes Theorem 4: it reviews bisection width and spectral expansion, re-derives the PST bisection-width Crossing Lemma with explicit constants, assembles Theorem 4, and records an Albertson-type corollary on regular spectral-expander critical graphs. Section 4 states the combined scope and limitations of the two observations, in particular that neither closes the Cranston residual triples at  $t \in \{25, 26\}$  and that a companion note [21] handles the orthogonal Fox–Pach–Suk sharpness question.

## 2 Observation 1: List-Albertson at chromatic number 18

### 2.1 Preliminaries on list-coloring

We work with finite simple graphs throughout. Standard graph-theory notation follows [13]. The *crossing number*  $\text{cr}(G)$  is the minimum number of pairwise edge-crossings over all drawings of  $G$  in the plane.

**Definition 5.** Let  $k \in \mathbb{N}$ . A graph  $G$  is *k-list-critical* (or *k-choice-critical*) if  $\chi_\ell(G) \geq k$  but  $\chi_\ell(G - e) < k$  for every edge  $e \in E(G)$ .

Equivalently,  $G$  is *k-list-critical* if and only if there exists a list assignment  $L$  with  $|L(v)| \geq k - 1$  for every vertex  $v$  such that  $G$  is not  $L$ -colourable but  $G - e$  is  $L$ -colourable for every edge  $e$ . Every graph with  $\chi_\ell(G) \geq k$  contains a *k-list-critical* subgraph (delete edges one by one until the  $\chi_\ell \geq k$  condition fails).

The following two lemmas are the only list-coloring inputs we use.

**Lemma 6** (List Dirac). *Every k-list-critical graph  $G$  satisfies  $\delta(G) \geq k - 1$ .*

*Proof.* Suppose not: let  $v \in V(G)$  have  $d_G(v) \leq k - 2$ . Let  $L$  be a list assignment with  $|L(u)| \geq k - 1$  for every  $u$  witnessing *k-list-criticality*, so  $G$  is not  $L$ -colourable but  $G - v$  is. Fix an  $L$ -colouring  $c$  of  $G - v$ . The neighbours of  $v$  in  $G$  use at most  $d_G(v) \leq k - 2$  colours from  $L(v)$ , leaving at least  $|L(v)| - d_G(v) \geq (k - 1) - (k - 2) = 1$  admissible colour for  $v$ . Extending  $c$  accordingly contradicts the non- $L$ -colourability of  $G$ .  $\square$

Lemma 6 is folklore in the list-coloring literature; see for instance Erdős–Rubin–Taylor [16, Theorem 2] for the same swap-a-colour argument. As an immediate consequence, every *k-list-critical* graph satisfies  $|E(G)| \geq \frac{k-1}{2}|V(G)|$ .

The list version of Brooks’ theorem characterising the equality case is also classical:

**Lemma 7** (List Brooks; Borodin [9], Erdős–Rubin–Taylor [16]). *Let  $G$  be a connected k-list-critical graph with  $\delta(G) = k - 1$ . Then either  $G \cong K_k$ , or  $k = 3$  and  $G$  is an odd cycle.*

The proof is a list-version of the original argument of Brooks, combined with a careful analysis of the Gallai-tree structure on the set of vertices of degree exactly  $k - 1$  (the *Gallai tree theorem* of Erdős–Rubin–Taylor and Borodin states that this set induces a graph each

of whose blocks is a complete graph or an odd cycle). We refer to [16, Theorem 3.2 and Theorem 3.5] and [9] for the details.

For our purposes only the immediate corollary matters: if a  $k$ -list-critical graph  $G$  has  $\delta(G) = k - 1$  and is not  $K_k$  (and is not an odd cycle when  $k = 3$ ), then in fact some vertex of  $G$  has degree  $\geq k$ , so  $|E(G)| \geq \frac{(k-1)|V(G)|+1}{2}$ . We do not need this sharper form for Theorem 3.

We will also use two standard topological / geometric inputs that have nothing to do with colour:

**Lemma 8** (Crossing Lemma; Ackerman [1]). *Let  $G$  be a graph drawn in the plane. Then*

$$\text{cr}(G) \geq \frac{|E(G)|^3}{29|V(G)|^2} \quad \text{whenever } |E(G)| \geq 7|V(G)|.$$

**Lemma 9** (Four-crossings-per-edge bound; Ackerman [1]). *If  $G$  admits a drawing in the plane in which every edge is crossed at most four times, then  $|E(G)| \leq 6|V(G)| - 12$ .*

Both lemmas are purely about drawings of abstract graphs; they make no reference to any colouring of  $G$ , and lift verbatim to the list-coloring setting. For the precise statements and attribution chain we follow Ackerman [1, Theorem 1.1 and Theorem 2.1]; the  $1/29$  constant of Lemma 8 is not the best one currently known (Bungener and Kaufmann [11] have  $1/27.48$  under a slightly larger edge-density hypothesis) but it is the constant under which the  $t \leq 18$  chain was originally closed and is sufficient for our purposes.

## 2.2 Proof of Theorem 3

We prove Theorem 3 by lifting each step of the Albertson–Cranston–Fox / Barát–Tóth / Ackerman chain to the list-coloring setting. The only ingredient that needs verification is that each step of the chain depends on  $\chi_\ell(G) \geq t$  *only through the edge floor on critical subgraphs*, and not through any property of vertex colourings that is specific to the non-list setting.

**Setup: reduce to a list-critical counterexample.** Fix  $t \leq 18$  and suppose  $G$  is a graph with  $\chi_\ell(G) \geq t$  but  $\text{cr}(G) < \text{cr}(K_t)$ . Since crossing number is monotone under edge-deletion ( $\text{cr}(G - e) \leq \text{cr}(G)$ ), passing to an edge-minimal subgraph with  $\chi_\ell \geq t$  preserves the inequality  $\text{cr}(G - e) \leq \text{cr}(G) < \text{cr}(K_t)$ . So we may assume  $G$  is  $t$ -list-critical.

By Lemma 6,  $\delta(G) \geq t - 1$ , hence

$$|E(G)| \geq \frac{t-1}{2}|V(G)|. \tag{3}$$

For  $t \geq 15$  this gives  $|E(G)| \geq 7|V(G)|$ , so the Crossing Lemma Lemma 8 applies (we will use a milder density hypothesis where convenient and treat the small- $t$  ranges separately).

### The Albertson–Cranston–Fox bound $|V(G)| \leq 4t$

We first record the lift of the basic ACF vertex bound. We work with Guy’s exact formula  $\text{cr}(K_t) \leq Z(t) := \frac{1}{4}\lfloor t/2 \rfloor \lfloor (t-1)/2 \rfloor \lfloor (t-2)/2 \rfloor \lfloor (t-3)/2 \rfloor$  as the upper bound on the right-hand side of the candidate counterexample inequality (Guy’s formula is conjectured to be sharp; this is irrelevant for us, only the upper bound matters); note  $Z(t) \leq t^4/64$ .

**Lemma 10** (List ACF bound). *If  $G$  is  $t$ -list-critical with  $\text{cr}(G) < \text{cr}(K_t)$  and  $t \geq 13$ , then  $|V(G)| \leq 4t$ .*

*Proof.* By Lemma 6 we have  $\delta(G) \geq t - 1$  and hence  $|E(G)| \geq \frac{t-1}{2}|V(G)|$ . For  $t \geq 13$  this gives  $|E(G)|/|V(G)| \geq 6 \geq \alpha$  for the constant  $\alpha = 4$  needed by Pach–Tóth’s Crossing Lemma [30] in the form  $\text{cr}(G) \geq |E|^3/(33.75|V|^2)$ . Combining,

$$\text{cr}(G) \geq \frac{(t-1)^3}{8 \cdot 33.75}|V(G)| = \frac{(t-1)^3}{270}|V(G)|.$$

On the other hand  $\text{cr}(G) < \text{cr}(K_t) \leq Z(t) \leq t^4/64$ , giving

$$\frac{(t-1)^3}{270}|V(G)| < \frac{t^4}{64}, \quad \text{i.e.} \quad |V(G)| < \frac{270}{64} \cdot \frac{t^4}{(t-1)^3} = \frac{135}{32} \cdot \frac{t^4}{(t-1)^3}.$$

For  $t \geq 13$  we have  $t/(t-1) \leq 13/12$  and  $(13/12)^3 \leq 1.275$ , so  $t^4/(t-1)^3 \leq 1.275t$  and  $|V(G)| < (135/32) \cdot 1.275t < 5.38t$ . The sharper Albertson–Cranston–Fox combinatorial bookkeeping in [4, §2], which depends only on the edge density and the Pach–Tóth Crossing Lemma, refines this to  $|V(G)| \leq 4t$ .  $\square$

The argument in [4, §2] uses, on the colouring side, only the bound  $\delta(G) \geq t - 1$  on the minimum degree of a  $t$ -critical MCE. By Lemma 6 the same bound holds for any  $t$ -list-critical MCE, and the rest of the ACF argument is a Crossing-Lemma manipulation that is insensitive to whether the criticality assumption is the chromatic-number version or the list-chromatic-number version. We therefore have the list-ACF vertex bound without further work.

### The Barát–Tóth refinement $|V(G)| \leq 3.57t$

Barát and Tóth [7, Theorem 1.2] improve the ACF bound to  $|V(G)| \leq 3.57t$  (and close Conjecture 1 for  $t \leq 16$ ) by replacing Pach–Tóth’s  $1/33.75$  Crossing-Lemma constant by the better  $1/31.1$  constant of Pach–Radoičić–Tardos–Tóth [31] and by a more careful accounting of the edge density on a minimum counterexample. Their accounting uses exactly the same colouring input as [4]:  $\delta(G) \geq t - 1$  for a  $t$ -critical MCE, used to drive  $|E(G)| \geq \frac{t-1}{2}|V(G)|$ .

**Lemma 11** (List Barát–Tóth bound). *If  $G$  is  $t$ -list-critical with  $\text{cr}(G) < \text{cr}(K_t)$  and  $t \geq 13$ , then  $|V(G)| \leq 3.57t$ .*

*Proof.* By Lemma 6, the edge floor (3) holds. The remainder of the proof of [7, Theorem 1.2] is a Crossing-Lemma argument that uses only this edge floor on the colouring side. Replacing “ $t$ -critical” by “ $t$ -list-critical” throughout the proof of [7, Theorem 1.2] yields the claim verbatim.  $\square$

### The Ackerman tightening $|V(G)| \leq 3.03t$

Ackerman [1] closes Conjecture 1 for  $t \leq 18$  by combining the Pach–Tóth–Tardos / Ackerman Crossing Lemma with the topological inequality Lemma 9. The overall scheme is: on a hypothetical MCE, an averaging argument forces a non-trivial fraction of edges to be crossed at most four times, which then triggers Lemma 9 on a subgraph.

**Lemma 12** (List Ackerman bound). *If  $G$  is  $t$ -list-critical with  $\text{cr}(G) < \text{cr}(K_t)$  and  $t \leq 18$  then  $|V(G)| \leq 3.03t$ .*

*Proof.* The proof of [1, Theorem 1] proceeds in three steps:

- (A) An edge floor on the MCE:  $|E(G)| \geq \frac{t-1}{2}|V(G)|$ .

- (B) A topological averaging argument: if the average number of crossings per edge in an optimal drawing of  $G$  is below an explicit threshold, a positive fraction of edges of  $G$  form a subgraph  $G'$  drawn with at most four crossings per edge.
- (C) An application of Lemma 9 to  $G'$ , combined with the edge floor (A) on  $G'$  inherited from (A) on  $G$ , to derive a contradiction unless  $|V(G)| \leq 3.03t$ .

Step (A) is the only step that uses any colouring hypothesis. The hypothesis used is precisely  $\delta(G) \geq t - 1$ , which by Lemma 6 holds for any  $t$ -list-critical  $G$ . Steps (B) and (C) are statements about *drawings* of  $G$  and *abstract subgraphs* of  $G$ ; neither references  $\chi$  or  $\chi_\ell$ . Substituting Lemma 6 for Dirac's theorem on critical graphs at step (A) and copying the proof of [1, Theorem 1] otherwise verbatim yields the stated bound.  $\square$

### Closing the small cases $t \leq 12$

For  $t \leq 12$ , the proof of Conjecture 1 in [4] combines the bound  $|V(G)| \leq 4t$  of Lemma 10 with a finite analysis of small graphs. Specifically, by Lemma 10 we may assume  $|V(G)| \leq 4t \leq 48$ , and by Lemma 6 we have  $\delta(G) \geq t - 1$ . For each  $t \in \{5, \dots, 12\}$ , the vertex range  $|V(G)| \leq 4t$  leaves only finitely many graphs to check, and each is either planar (so  $\text{cr}(G) = 0$  which forces  $\chi_\ell(G) \leq 5$  by Thomassen's theorem [34], contradicting  $t \geq 6$ ), or it contains  $K_t$  as a subgraph (in which case  $\text{cr}(G) \geq \text{cr}(K_t)$  by the monotonicity of crossing number), or it falls into the finitely many residual configurations handled hand-by-hand in [4].

For  $t \leq 4$ , Conjecture 2 reduces to elementary observations: a graph with  $\chi_\ell(G) \geq 2$  has at least one edge, so  $\text{cr}(G) \geq 0 = \text{cr}(K_2)$ ; a graph with  $\chi_\ell(G) \geq 3$  is non-planar in the relevant sense via the list-coloring version of the Four Colour Theorem for the planar case (every planar graph is 5-choosable [34], so  $\chi_\ell(G) \geq 3$  does not by itself force  $\text{cr}(G) > 0$ , but  $\text{cr}(K_3) = 0$  so the bound is trivial); the case  $t = 4$  uses  $\text{cr}(K_4) = 0$  and again the inequality holds trivially. The list-coloring versions of Brooks' theorem and of Vizing's theorem on  $\chi_\ell$  for  $t = 4$  are by now classical (Vizing [35]; Erdős–Rubin–Taylor [16]; Borodin [9]); we omit further detail.

### Assembly

*Proof of Theorem 3.* Suppose for contradiction that  $G$  is a counterexample to Theorem 3, that is  $\chi_\ell(G) \geq t$  and  $\text{cr}(G) < \text{cr}(K_t)$  for some  $t \leq 18$ .

For  $t \leq 12$  the claim follows from the small-case analysis above applied to an edge-minimal subgraph of  $G$  with  $\chi_\ell \geq t$ .

For  $13 \leq t \leq 18$ , after passing to a  $t$ -list-critical subgraph (which preserves both hypotheses by the monotonicity arguments above), Lemma 12 forces  $|V(G)| \leq 3.03t$ . The proof of [1, Theorem 1] now combines Lemma 9, the Pach–Tóth–Tardos–Tóth Crossing Lemma, and the edge floor (3) to derive  $\text{cr}(G) \geq \text{cr}(K_t)$  on every  $G$  with  $|V(G)| \leq 3.03t$  and  $\delta(G) \geq t - 1$ ; the colouring hypothesis enters only via the latter inequality and the corresponding edge floor, both of which we have. The argument therefore goes through verbatim with “ $t$ -critical” replaced by “ $t$ -list-critical”, contradicting  $\text{cr}(G) < \text{cr}(K_t)$ .  $\square$

**Remark on the Brooks/Gallai-tree step.** The careful reader will notice that we did not in fact use the list-Brooks theorem (Lemma 7). The reason is that the Albertson chain of [4, 7, 1] uses only the *average* edge floor  $|E(G)| \geq \frac{t-1}{2}|V(G)|$ , which follows from  $\delta(G) \geq t - 1$  alone (Lemma 6). The stronger Brooks/Gallai-tree input is needed only to rule out the equality case  $|E(G)| = \frac{t-1}{2}|V(G)|$ , which corresponds to  $G$  being  $K_t$  (and we already have  $\text{cr}(K_t) \geq \text{cr}(K_t)$  for  $G = K_t$  trivially) or, for  $t = 3$ , an odd cycle. We state

Lemma 7 explicitly in Section 2.1 because it confirms that the rare critical case of the equality  $|E(G)| = \frac{t-1}{2}|V(G)|$  presents no additional obstacle.

### 2.3 Why $t \leq 18$ and not $t \leq 24$

The boundary  $t = 18$  in Theorem 3 is inherited directly from Ackerman’s threshold for the ordinary statement of Albertson. Cranston [12] has very recently extended Conjecture 1 to  $t \leq 24$  (with a small residual set of triples  $(t, n) \in \{(25, 48), (26, 50), (26, 51)\}$ ) by combining the same edge-floor plus Crossing-Lemma analysis with the new weak-immersion vertex bound of Fox–Pach–Suk [17, Theorem 1.2(i)].

This subsection records why the same extension does *not* currently carry over to the list-coloring setting. The obstruction is sharp and lies in a single auxiliary lemma.

**The Fox–Pach–Suk vertex bound.** The relevant statement is [17, Theorem 1.2(i)]: every graph  $G$  with  $\chi(G) \geq k$  and  $|V(G)| < 1.4(k - 1)$  contains a weak immersion of  $K_k$ . The proof of [17, Theorem 1.2(i)] uses an auxiliary multigraph  $H$  constructed from a Gallai decomposition of  $G$  via an optimal proper vertex colouring, and bounds its chromatic index by

$$\chi'(H) \leq (9/16 + o(1))k, \quad (4)$$

which is [17, Lemma 2.3]. The constant  $9/16$  is sharp within their Vizing–Gupta + semi-random framework; we have shown in a companion note [21] that any improvement of  $9/16$  requires modifying that framework, not merely re-tuning its parameters.

**The list-edge-coloring analogue.** The list-coloring lift of the Cranston extension to  $t \leq 24$  requires the list-edge-colouring analogue of (4) at the same constant:

$$\chi'_\ell(H) \leq (c_\ell + o(1))k \quad (5)$$

with  $c_\ell = 9/16$ , applied to the same multigraph class as in [17, Lemma 2.3]. The best published bound of this form is due to Borodin–Kostochka–Woodall [10], who prove

$$\chi'_\ell(H) \leq \frac{7}{4}\Delta(H) + O(1)$$

for multigraphs  $H$ . Substituting their  $c_\ell = 7/4$  in place of the  $9/16$  chromatic-index bound and propagating the resulting loss through the Fox–Pach–Suk argument degrades their vertex bound  $|V(G)| < 1.4(k - 1)$  to roughly

$$|V(G)| < 1.4 \cdot \frac{9/16}{7/4} \cdot (k - 1) = \frac{1.4 \cdot 9}{28}(k - 1) = 0.45(k - 1),$$

a factor  $\approx 3.1$  smaller than the constant  $1.4$  used by Cranston. The corresponding list-version of the Cranston chain *cannot reach past the Ackerman threshold*  $t = 18$  with the currently published constants on the right-hand side of (5).

We state the corresponding conditional theorem; we do not claim its hypothesis is reachable.

**Theorem 13** (Conditional). *Suppose that (5) holds at  $c_\ell = 9/16$  for the multigraphs  $H$  constructed in [17, §3] from an optimal list-colouring of the underlying graph  $G$ , in place of an optimal proper vertex colouring. Then for every  $t \leq 24$ , every graph  $G$  with  $\chi_\ell(G) \geq t$  satisfies  $\text{cr}(G) \geq \text{cr}(K_t)$ .*



*Proof sketch.* Repeat the argument of [12], replacing every invocation of  $\chi$  by  $\chi_\ell$  and every invocation of [17, Lemma 2.3] at constant  $9/16$  by the hypothesised list-edge-colouring analogue (5) at the same constant  $9/16$ . The ingredients on the Crossing-Lemma side – the Bungenier–Kaufmann [11] Crossing-Lemma constant  $1/27.48$  and the corresponding edge-density threshold – are purely topological and lift verbatim. The only place where the substitution non-trivially affects the conclusion is at the chromatic-index step (5), which is by hypothesis available at the original constant  $9/16$ .  $\square$

**Remark 14.** Theorem 13 should be read as a precise statement of the gap between the ordinary and the list versions of the Cranston extension; we are not aware of any approach to its hypothesis that does not require new results on list-edge-colouring of multigraphs at the Goldberg–Seymour boundary. The Borodin–Kostochka–Woodall constant  $7/4$  has not been improved in the intervening decades.

## 2.4 DP-coloring corollary

The *DP-chromatic number* (or *correspondence-chromatic number*)  $\chi_{\text{DP}}(G)$  introduced by Dvořák and Postle [15] satisfies  $\chi_{\text{DP}}(G) \geq \chi_\ell(G)$  for every graph  $G$ . The proof of Theorem 3 did not use any specific property of list-coloring beyond the DP-lift of Lemma 6: every  $k$ -DP-critical graph has minimum degree at least  $k - 1$ , by the verbatim swap argument of Lemma 6 applied to a DP-cover (see Bernshteyn–Kostochka–Pron [8]). Hence we have:

**Corollary 15.** *Let  $G$  be a graph and let  $t \leq 18$ . If  $\chi_{\text{DP}}(G) \geq t$ , then  $\text{cr}(G) \geq \text{cr}(K_t)$ .*

*Proof.* Repeat the proof of Theorem 3, replacing Lemma 6 by the corresponding DP-Dirac statement and “ $t$ -list-critical” by “ $t$ -DP-critical” throughout.  $\square$

## 3 Observation 2: A bisection-width Crossing Lemma for regular spectral expanders

### 3.1 Preliminaries on bisection width and spectral expansion

**Definition 16** (Bisection width). The *bisection width* of a graph  $G$  on  $n$  vertices is

$$\text{bw}(G) := \min\{|E(A, B)| : V(G) = A \sqcup B, \lfloor n/2 \rfloor \leq |A|, |B| \leq \lceil n/2 \rceil\},$$

where  $E(A, B)$  denotes the set of edges of  $G$  with one endpoint in  $A$  and one in  $B$ .

We follow the convention of Pach–Spencer–Tóth [29] and require that the bipartition be *balanced*. Some authors (e.g. [24]) instead require only that  $|A|, |B| \geq n/3$ ; the two conventions differ by at most a factor of 2 in the resulting  $\text{bw}(G)$  on regular graphs, which is absorbed into the implicit constants below.

**Definition 17** (Spectral expansion). Let  $G$  be a  $d$ -regular graph on  $n$  vertices and let  $A_G \in \mathbb{R}^{n \times n}$  be its adjacency matrix. The eigenvalues of  $A_G$  are real,  $d = \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n \geq -d$ . We write  $\lambda_2(G) := \max(|\lambda_2|, |\lambda_n|)$  for the *second adjacency eigenvalue*, and call  $G$  a  $\theta$ -*spectral expander* (for  $\theta \in [0, 1)$ ) if  $\lambda_2(G) \leq \theta d$ .

The fundamental tool relating spectrum to combinatorics is the expander mixing lemma:

**Lemma 18** (Expander mixing lemma; Alon–Chung [6]). *Let  $G$  be a  $d$ -regular graph on  $n$  vertices with second adjacency eigenvalue  $\lambda_2(G) \leq \theta d$ . For every pair of disjoint sets  $S, T \subseteq V(G)$ ,*

$$\left| e(S, T) - \frac{d|S||T|}{n} \right| \leq \theta d \sqrt{|S||T| \left(1 - |S|/n\right) \left(1 - |T|/n\right)}. \quad (6)$$

**Corollary 19** (Spectral bisection bound). *Let  $G$  be a  $d$ -regular graph on  $n$  vertices with  $\lambda_2(G) \leq \theta d$ . Then*

$$\text{bw}(G) \geq \frac{(1 - \theta) d n}{4}.$$

*Proof.* Let  $V(G) = A \sqcup B$  be a balanced bipartition. Suppose first that  $n$  is even, so  $|A| = |B| = n/2$ . By Lemma 18 with  $S = A$ ,  $T = B$ ,  $|S| = |T| = n/2$  and  $1 - |S|/n = 1/2$ ,

$$e(A, B) \geq \frac{d(n/2)(n/2)}{n} - \theta d \sqrt{(n/2)(n/2)(1/2)(1/2)} = \frac{d n}{4} - \theta d \cdot \frac{n}{4} = \frac{(1 - \theta) d n}{4}.$$

Taking the minimum over balanced bipartitions gives the claim for  $n$  even. For  $n$  odd the same calculation with  $|S| = \lfloor n/2 \rfloor$  and  $|T| = \lceil n/2 \rceil$  gives  $e(A, B) \geq (1 - \theta) d (n^2 - 1)/(4n) \geq (1 - \theta) d n/4 \cdot (1 - 1/n^2)$ , which differs from the stated bound by  $O(d/n)$ , absorbed into the inequality for  $n \geq 2$ .  $\square$

Corollary 19 is folklore. The constant  $1/4$  is sharp at  $\theta = 0$  (e.g. on  $K_{n,n}$ ,  $d = n/2$ ,  $\lambda_2 = 0$ , and  $\text{bw}(K_{n,n}) = n^2/4 = d n/2$ , exceeding the bound). For explicit references, see [19, Theorem 2.4] for a textbook account of the expander mixing lemma and its bisection-width consequence.

**Lemma 20** (Ramanujan bound; Alon–Boppana [5, 26]). *Every  $d$ -regular graph on  $n$  vertices satisfies  $\lambda_2(G) \geq 2\sqrt{d-1} - o_n(1)$ . Equivalently, the best possible spectral expansion of an infinite family of  $d$ -regular graphs is  $\theta = 2\sqrt{d-1}/d$ . Graphs achieving this bound asymptotically are called Ramanujan graphs; explicit constructions exist for  $d = p + 1$  with  $p$  prime [25].*

### 3.2 A self-contained bisection-width Crossing Lemma

We derive the PST inequality (1) inline from the dual bisection inequality of Pach–Shahrokhi–Szegedy and Sýkora–Vrt’o, in its conservative  $1/80$  form.

**Lemma 21** (Bisection-width Crossing Lemma; conservative form). *Every graph  $G$  on  $n$  vertices satisfies*

$$\text{cr}(G) \geq \frac{\text{bw}(G)^2}{80} - \frac{\sum_{v \in V(G)} \deg(v)^2}{16}. \quad (7)$$

*Proof.* Pach–Shahrokhi–Szegedy [28] and Sýkora–Vrt’o [33] independently established the dual estimate

$$\text{bw}(G) \leq 6.32 \sqrt{\text{cr}(G)} + 1.58 \sqrt{\sum_v \deg(v)^2}. \quad (8)$$

Squaring and using  $(a + b)^2 \leq 2a^2 + 2b^2$  on the right-hand side gives

$$\text{bw}(G)^2 \leq 2 \cdot (6.32)^2 \text{cr}(G) + 2 \cdot (1.58)^2 \sum_v \deg(v)^2 \leq 79.9 \text{cr}(G) + 5.00 \sum_v \deg(v)^2.$$

Rearranging and using the bookkeeping bounds  $79.9 \leq 80$  and  $5.00/79.9 \leq 1/16$ ,

$$\text{cr}(G) \geq \frac{\text{bw}(G)^2}{79.9} - \frac{5.00}{79.9} \cdot \sum_v \deg(v)^2 \geq \frac{\text{bw}(G)^2}{80} - \frac{\sum_v \deg(v)^2}{16}. \quad \square$$

**Remark 22.** The conservative proof above gives the constant  $1/80$  on the  $\text{bw}(G)^2$  term. Pach, Spencer, and Tóth [29, Section 4] sharpen this to  $1/40$  by a more careful split of the dual inequality (8) – allocating the two right-hand-side contributions optimally before squaring, rather than using the  $(a + b)^2 \leq 2a^2 + 2b^2$  bookkeeping step in the proof above. Their proof yields

$$\text{cr}(G) \geq \frac{\text{bw}(G)^2}{40} - \frac{\sum_v \deg(v)^2}{16}.$$

Substituting this sharper  $1/40$  in place of  $1/80$  everywhere in the proof of Theorem 4 replaces the constant  $1/1280$  by  $1/640$  in (2), doubling the leading expander coefficient. We work with the self-contained  $1/80$  form throughout the remainder of this note.

### 3.3 Proof of Theorem 4 and numerical illustration

*Proof of Theorem 4.* Let  $G$  be  $d_0$ -regular on  $n$  vertices with  $\lambda_2(G) \leq \theta d_0$ . By Lemma 21,

$$\text{cr}(G) \geq \frac{\text{bw}(G)^2}{80} - \frac{\sum_v \deg(v)^2}{16} = \frac{\text{bw}(G)^2}{80} - \frac{d_0^2 n}{16},$$

where the last equality uses  $\sum_v \deg(v)^2 = d_0^2 n$  for  $d_0$ -regular  $G$ . By Corollary 19,  $\text{bw}(G) \geq (1 - \theta) d_0 n/4$ , so  $\text{bw}(G)^2 \geq (1 - \theta)^2 d_0^2 n^2/16$ . Substituting,

$$\text{cr}(G) \geq \frac{1}{80} \cdot \frac{(1 - \theta)^2 d_0^2 n^2}{16} - \frac{d_0^2 n}{16} = \frac{(1 - \theta)^2 d_0^2 n^2}{1280} - \frac{d_0^2 n}{16}. \quad \square$$

**Numerical sanity check at a Ramanujan regime.** We illustrate the strength of Theorem 4 at one concrete spectral-expander regime. Take  $d_0 = 10$ ,  $n = 100$ , and let  $G$  be a 10-regular Ramanujan graph (so  $\lambda_2(G) = 2\sqrt{9} = 6$ , giving  $\theta = 0.6$ ). Then  $m = d_0 n/2 = 500$  edges, and:

- Theorem 4 gives  $\text{cr}(G) \geq (0.4)^2 \cdot 100 \cdot 10000/1280 - 100 \cdot 100/16 = 16 \cdot 10000/1280 - 625 = 125 - 625$ , which is *negative*; the bound is vacuous at  $n = 100$ .
- Doubling to  $n = 200$  ( $d_0 = 10$ ,  $\theta = 0.6$ , Ramanujan): Theorem 4 gives  $0.16 \cdot 100 \cdot 40000/1280 - 100 \cdot 200/16 = 500 - 1250 < 0$  – still vacuous at this  $\theta$ .
- At  $n = 400$ ,  $d_0 = 10$ ,  $\theta = 0.6$ :  $0.16 \cdot 100 \cdot 160000/1280 - 100 \cdot 400/16 = 2000 - 2500 < 0$  – still vacuous.
- At  $n = 1000$ ,  $d_0 = 10$ ,  $\theta = 0.6$ :  $0.16 \cdot 100 \cdot 10^6/1280 - 100 \cdot 1000/16 = 12500 - 6250 = 6250$ ; positive.
- The Bungenier–Kaufmann Crossing Lemma at the same  $(n, d_0) = (1000, 10)$  gives  $m = 5000$  and  $\text{cr}(G) \geq m^3/(27.48 n^2) \approx 1.25 \cdot 10^{11}/(2.748 \cdot 10^7) \approx 4550$ .

At  $(n, d_0, \theta) = (1000, 10, 0.6)$ , the headline conservative form of Theorem 4 is roughly  $1.4\times$  stronger than BK; using the sharper PST  $1/40$  constant (Remark 22) doubles the leading term and roughly triples the gap. The improvement grows as  $d_0$  grows (because the bisection bound scales as  $d_0 n$ , giving  $\text{cr} \gtrsim d_0^2 n^2$ , while the BK bound on a  $d_0$ -regular graph gives  $\text{cr} \gtrsim d_0^3 n$ , smaller by a factor  $n/d_0$ ).

**Density threshold.** The bound (2) of Theorem 4 is non-trivial (positive right-hand side) exactly when

$$\frac{(1 - \theta)^2 d_0^2 n^2}{1280} > \frac{d_0^2 n}{16}, \quad \text{i.e.,} \quad n > \frac{80}{(1 - \theta)^2}.$$

For  $\theta = 0.6$  this gives  $n > 500$ ; for  $\theta = 0$  (complete-bipartite-like spectrum),  $n > 80$ . Below this threshold the second term dominates and the inequality is vacuous. Using the sharper PST  $1/40$  constant (Remark 22) halves these thresholds.

### 3.4 The Albertson corollary

We now record an Albertson-type corollary of Theorem 4 on the class of  $t$ -critical  $d_0$ -regular spectral expanders.

**Setup.** Recall Guy's formula [18] for the conjectured crossing number of the complete graph  $K_t$ ,

$$\text{cr}(K_t) \leq Z(t) := \frac{1}{4} \lfloor t/2 \rfloor \lfloor (t-1)/2 \rfloor \lfloor (t-2)/2 \rfloor \lfloor (t-3)/2 \rfloor.$$

The conjecture is open for  $t \geq 13$ , but the upper bound  $\text{cr}(K_t) \leq Z(t)$  is unconditional (realised by Zarankiewicz's drawing). We have  $Z(t) \leq t^4/64$ .

Recall also Dirac's theorem [14]: every  $t$ -critical graph  $G$  satisfies  $\delta(G) \geq t-1$ . The corresponding list-critical bound (Lemma 6 in Section 2) gives the same conclusion for  $t$ -list-critical  $G$ .

**Corollary 23** (Albertson for regular spectral-expander critical graphs). *Let  $t \geq 4$  and let  $G$  be a  $t$ -critical (or  $t$ -list-critical)  $d_0$ -regular graph on  $n$  vertices with  $\lambda_2(G) \leq \theta d_0$ . Suppose*

(a)  $d_0 \geq t-1$  (automatic from the Dirac bound on any  $t$ -critical graph),

(b)  $(1-\theta)^2 \geq 1280(Z(t) + d_0^2 n/16)/(d_0^2 n^2)$ , equivalently

$$(1-\theta)^2 d_0^2 n^2 \geq 1280 Z(t) + 80 d_0^2 n. \quad (9)$$

Then  $\text{cr}(G) \geq \text{cr}(K_t)$ .

*Proof.* Theorem 4 gives  $\text{cr}(G) \geq (1-\theta)^2 d_0^2 n^2/1280 - d_0^2 n/16$ . Hypothesis (b) is exactly the inequality

$$\frac{(1-\theta)^2 d_0^2 n^2}{1280} \geq Z(t) + \frac{d_0^2 n}{16},$$

which rearranges to  $\text{cr}(G) \geq Z(t) \geq \text{cr}(K_t)$ .  $\square$

**Simplified asymptotic form.** For large  $t$  and large  $n$ , condition (9) simplifies. Using  $Z(t) \leq t^4/64$  and  $d_0 \geq t-1 \geq t/2$  (the latter for  $t \geq 2$ ),

$$\frac{1280 Z(t)}{d_0^2 n^2} \leq \frac{1280 t^4/64}{(t/2)^2 n^2} = \frac{80 t^2}{n^2},$$

so (9) is implied by  $(1-\theta)^2 \geq 80 t^2/n^2 + 80/n$ , equivalently

$$\theta \leq 1 - \sqrt{80 t^2/n^2 + 80/n}. \quad (10)$$

For  $n \gtrsim 10t$ , the term  $80 t^2/n^2 \leq 80/100 = 0.80$  already gives  $\theta \leq 1 - 0.90 \approx 0.10$ ; for  $n \gtrsim 20t$ , the bound becomes  $\theta \leq 1 - 0.32$ , so any  $\theta \leq 0.68$  suffices. This is well within Ramanujan range  $\theta = 2\sqrt{d_0-1}/d_0$  for  $d_0 \geq 10$  (where  $\theta_{\text{Ramanujan}} \approx 0.6$ ).

**Smallest  $t$  where the corollary is non-vacuous.** For  $t = 4$ , we have  $\text{cr}(K_4) = 0$ , so the inequality is trivial. For  $t = 5$ ,  $\text{cr}(K_5) = 1$ , also easily met. The first non-trivial case is  $t = 7$  ( $\text{cr}(K_7) = 9$ ), or  $t = 13$  if one demands  $\text{cr}(K_t)$  matching  $Z(t)$  unconditionally (the regime where Albertson is genuinely open). For  $t = 13$ ,  $Z(13) = 225$ , so (9) requires  $(1-\theta)^2 d_0^2 n^2 \geq 288000 + 80 d_0^2 n$ . At  $d_0 = 12$  and  $n = 60$ , the left-hand side is  $(1-\theta)^2 \cdot 144 \cdot 3600 = 518400(1-\theta)^2$  and the right-hand side is  $288000 + 80 \cdot 144 \cdot 60 = 288000 + 691200 = 979200$ , exceeding the left-hand side even at  $\theta = 0$ . At  $d_0 = 12$  and  $n = 120$  the left-hand side is  $(1-\theta)^2 \cdot 144 \cdot 14400 = 2073600(1-\theta)^2$  and the right-hand side is  $288000 + 80 \cdot 144 \cdot 120 = 288000 + 1382400 = 1670400$ , giving  $(1-\theta)^2 \geq 1670400/2073600 \approx 0.806$ , i.e.  $\theta \leq 1 - 0.898 \approx 0.102$ . The Ramanujan bound for  $d_0 = 12$  is  $\theta = 2\sqrt{11}/12 \approx 0.553$ , so even optimal expanders at this regime do not meet the corollary's hypothesis under the conservative  $1/1280$  constant; using the sharper PST  $1/40$  form (Remark 22) halves the coefficients in (9) and brings the threshold down

to  $\theta \leq 0.398$ , which a Ramanujan graph at  $d_0 = 12, n = 120$  already meets. The corollary becomes non-vacuous for Ramanujan expanders once  $n \gtrsim 30t$  under the conservative form ( $n \gtrsim 15t$  under the sharper PST form), which is well above the Cranston regime  $n \approx 2t$ .

This is the principal honest qualification of Theorem 4: the corollary produces a Crossing-Lemma improvement on *moderately dense* expander critical graphs ( $n \gtrsim 20t$ ), not on the tight Dirac-edge-floor regime ( $n \approx 2t$ ) that the Albertson conjecture is concerned with. The Ore-type critical-graph candidates populating the Cranston residual in particular fall well outside the spectral-expander class; we elaborate in Section 4.

## 4 Common scope and limitations

### 4.1 The Cranston residual is untouched by either observation

The triples  $(t, n) \in \{(25, 48), (26, 50), (26, 51)\}$  at which the unconditional Albertson conjecture is currently open [12] are populated by Ore compositions of  $K_{26}$  [27]: the construction takes two copies of  $K_{26}$  joined by a small “interface” via edge-identification on a small common subgraph. The resulting graph is 25-regular on  $n = 50$  or  $51$  vertices, with edge count meeting the Kostochka–Yancey [20] bound  $m \geq \lceil ((t+1)(t-2)n - t(t-3))/(2(t-1)) \rceil = 649$  at  $(t, n) = (26, 51)$ .

**Why Theorem 3 does not apply.** The Cranston residual lies in the range  $t \in \{25, 26\}$ , which is strictly above the Ackerman threshold  $t \leq 18$  inherited by Theorem 3. A list version of Albertson at  $t \in \{25, 26\}$  would in particular imply the corresponding ordinary case via the trivial list assignment  $L(v) = \{1, \dots, \chi(G)\}$ , but this is precisely the original  $t \in \{25, 26\}$  problem; closing the list case at those thresholds is therefore not easier than closing Cranston’s residual.

**Why Theorem 4 does not apply: Ore graphs are not spectral expanders.** The adjacency matrix of an Ore composition  $G$  has a clear block-bipartition structure: viewing  $V(G)$  as  $A \sqcup B$  with  $|A| = |B| = 25$  or  $26$ , the within-block edges (each block is nearly  $K_{26}$ ) contribute eigenvalues bounded by the  $K_{26}$  spectrum, but the small inter-block interface produces a near-zero second eigenvalue *only relative to the  $K_{n,n}$ -like structure*. Concretely, a near-balanced bipartition of  $G$  via  $(A, B)$  has very few cut edges (the  $\Theta(1)$  interface), so

$$\text{bw}(G) = O(1) \quad (\text{not } \Omega(d_0 n) \text{ as for expanders}).$$

By Corollary 19 in reverse, this forces  $\theta$  to be *very close to 1*: if  $\text{bw}(G) \leq c$  with  $c = O(1)$ , then  $(1 - \theta) d_0 n / 4 \leq c$ , giving  $1 - \theta \leq 4c / (d_0 n) = O(1/n)$ . Hence  $\theta = 1 - O(1/n)$ , and the constant  $(1 - \theta)^2$  in (2) is  $O(1/n^2)$ , making the bound vacuous.

In particular, applying Theorem 4 to the Ore corner graphs  $(n, d_0) = (51, 25)$  with  $\theta = 1 - O(1/51)$  gives  $\text{cr}(G) \geq O(1) - 25^2 \cdot 51/16 = O(1) - 1992$ , which is not useful.

### 4.2 Open: non-expander reduction

The natural further question is whether any  $t$ -critical non-expander can be  $\varepsilon$ -perturbed into an expander preserving criticality. Possibilities:

- *Cayley-graph cover.* Replace  $G$  by a Cayley graph  $\Gamma$  on a group  $\Gamma$  acting on  $V(G)$ , with the generating set chosen so that  $\Gamma$  is a spectral expander and admits  $G$  as a quotient. If  $\Gamma$  inherits  $t$ -criticality from  $G$ , Theorem 4 applies to  $\Gamma$ , giving  $\text{cr}(\Gamma) \geq \text{explicit}$ . By the standard minor-monotonicity of crossing number,  $\text{cr}(\Gamma) \geq \text{cr}(G)$  on subgraphs, but

the  $\varepsilon$ -perturbation goes in the wrong direction here (adding edges to  $G$  to produce  $\Gamma$  does not give  $\text{cr}(G) \geq \text{cr}(\Gamma)$ ).

- *Lifted-product construction.* The recent lifted-product constructions of quantum LDPC codes [32] produce explicit expanders from non-expander inputs by a topological lift. The applicability to preserve  $t$ -criticality is not, to our knowledge, addressed in the literature.

We do not pursue these directions here.

### 4.3 Relation to the companion sharpness note

The companion note [21] addresses an orthogonal structural slack in the same Albertson chain: the sharpness of the Fox–Pach–Suk constant  $9/16$  in their [17, Lemma 2.3]. Specifically, [21] establishes that any improvement of  $9/16$  requires changing the Vizing–Gupta plus semi-random framework rather than re-tuning its degree-threshold parameter  $\delta = 9/8$ . This places a sharp boundary on how much the ordinary Cranston extension to  $t \leq 24$  can itself be improved; the present note records two parallel boundaries (the chromatic-vs-list-chromatic boundary in Section 2 and the density-vs-expander boundary in Section 3), neither of which closes the Cranston residual either.

Together, the two notes record three identifiable structural slacks in the Albertson chain:

- the FPS chromatic-index constant ( $9/16$ , sharpness established in [21]);
- the chromatic-versus-list-chromatic boundary at  $t \leq 18$  (Theorem 3);
- the density-vs-expander boundary in the Crossing-Lemma constant (Theorem 4).

None of the three closes the Cranston residual at  $t \in \{25, 26\}$ ; each is a stand-alone graph-theory observation in the structural neighbourhood of Conjecture 1.

## Acknowledgements

This paper assembles two side-observations that emerged while investigating Albertson’s conjecture. The list-coloring observation (Theorem 3) arose from inspecting the ingredients of the unconditional Albertson chain up to  $t = 18$  and noticing that none of them uses the chromatic-number hypothesis past the Dirac minimum-degree bound. The spectral-expander observation (Theorem 4) emerged from an attempt at a direct min-degree-aware Crossing Lemma that failed cleanly: the density-iteration machinery of all known Crossing-Lemma proofs is degree-sequence insensitive at its iteration-stopping step, and the present packaging of PST plus Alon is the proven fallback. We thank our collaborators for many useful discussions on the Albertson conjecture, on critical-graph theory, on the list-Brooks theorem, and on the spectral methods of expander graph theory.

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